Supplementary Material for "DiMA-PPI: Thermodynamically Integrable State-Potential Learning for Protein–Protein Binding Mutations"

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1. Additional benchmark generalization

Table S1: Additional historical pairwise-antisymmetry benchmarks. Reported baselines use the DSSA-PPI publication. The local official-input variant uses the released 15 Å interface graphs and chain-level ESM-2 features; full-input variants use reconstructed partner structures and residue-level multimodal features. These experiments predate the canonical state-potential formulation. Local values are mean ± standard deviation over seeds 42–46. Bold marks the better locally controlled full-input formulation, not a statistical comparison with DSSA-PPI.

Dataset	Method	N	Pearson	AUROC
S4169	DSSA-PPI (reported)	–	0.58 ± 0.01	0.70 ± 0.02
	Soft pairwise (official-input)	3920	0.514 ± 0.053	0.678 ± 0.010
	Soft pairwise (full-input)	4160	0.579 ± 0.019	0.700 ± 0.004
	Hard pairwise (full-input)	4160	$\mathbf{0.603 \pm 0.015}$	$\mathbf{0.712 \pm 0.006}$
M1707	DSSA-PPI (reported)	–	0.42 ± 0.02	0.72 ± 0.02
	Soft pairwise (official-input)	1112	0.382 ± 0.019	0.705 ± 0.020
	Soft pairwise (full-input)	1337	0.411 ± 0.021	0.690 ± 0.022
	Hard pairwise (full-input)	1337	$\mathbf{0.454 \pm 0.029}$	$\mathbf{0.736 \pm 0.022}$

The official-input and full-input rows retain different observation cohorts, and DSSA-PPI additionally uses separate pretraining. Cross-row contrasts involving these pipelines are therefore descriptive. The soft-versus-hard full-input rows use the same observations and provide the locally controlled comparison.

2. Complete real-state integrability audit

The audit uses 92 held-out mutation panels and representative seed 43. It contains 5,653 reverse comparisons, 9,904 ordered three-state paths, 31,922 deterministic triangle cycles, and 133,736 deterministic four-state cycles per model. Exactness of the state-potential construction is seed independent. The intermediate mutant-to-mutant edges are not independently labelled; these statistics measure internal thermodynamic consistency rather than the predictive accuracy of those edges.

Table S2: Complete reverse, path, and cycle audit. Values are in kcal mol^{-1}. "Below" columns give the fraction of absolute errors below the stated threshold.

Model	Relation	Mean	Median	P95	Maximum	Below 0.1	Below 10^{-3}
Pairwise	Reverse	2.295	1.264	5.540	21.600	0.047	0.0005
	Path	1.333	0.758	3.537	16.742	0.088	0.0007
	Triangle	3.287	1.812	7.915	32.330	0.038	0.0003
	Four-state cycle	4.209	2.359	10.335	43.129	0.028	0.0002
Soft consistency	Reverse	0.892	0.581	2.746	7.073	0.108	0.0014
	Path	0.920	0.549	2.952	9.111	0.120	0.0016
	Triangle	1.346	0.852	4.180	10.595	0.080	0.0010
	Four-state cycle	1.709	1.080	5.292	15.341	0.064	0.0005
Hard antisymmetry	Reverse	1.18×10^{-7}	5.96×10^{-8}	4.77×10^{-7}	4.29×10^{-6}	1.000	1.000
	Path	0.417	0.215	1.478	5.765	0.300	0.0036
	Triangle	0.276	0.154	0.955	5.765	0.376	0.0048
	Four-state cycle	0.315	0.177	1.078	7.343	0.331	0.0039
Canonical state potential	Reverse	1.84×10^{-7}	1.19×10^{-7}	5.96×10^{-7}	2.86×10^{-6}	1.000	1.000
	Path	3.12×10^{-7}	1.79×10^{-7}	1.06×10^{-6}	4.71×10^{-6}	1.000	1.000
	Triangle	2.50×10^{-7}	1.79×10^{-7}	7.45×10^{-7}	3.76×10^{-6}	1.000	1.000
	Four-state cycle	2.91×10^{-7}	2.09×10^{-7}	8.59×10^{-7}	3.81×10^{-6}	1.000	1.000

3. Complete distribution-shift results

Table S3: Five-seed ensemble generalization under sequence- and interface-structure-cluster folds. The two partitions probe parallel novelty axes.

Split	Model	Pearson	Spearman	RMSE	MAE	AUROC
Sequence cluster	Unconstrained backbone	0.341	0.431	2.290	1.520	0.684
	Soft consistency	0.380	0.451	2.252	1.502	0.690
	Hard antisymmetry	0.392	0.475	2.257	1.487	0.733
	Canonical state potential	0.354	0.441	2.355	1.519	0.743
Interface structure	Unconstrained backbone	0.469	0.547	2.163	1.451	0.767
	Soft consistency	0.490	0.552	2.136	1.421	0.777
	Hard antisymmetry	0.463	0.509	2.194	1.463	0.766
	Canonical state potential	0.488	0.515	2.174	1.458	0.771

All complex-bootstrap intervals for state-potential minus backbone, soft, or hard crossed zero for PCC, Spearman, RMSE, and MAE on both partitions. The sequence-split AUROC differences versus backbone and soft were positive with intervals excluding zero, but AUROC is complementary and the primary continuous regression metrics do not support a generalization claim. Complete machine-readable intervals are supplied in the reproducibility archive.

Table S4: External evaluation on 395 mutations from 16 sequence-decontaminated AbBind complexes. Values are mean ± standard deviation over five independently trained seeds. Complex-bootstrap intervals crossed zero for every model difference.

Model	Pearson	Spearman	RMSE	MAE	AUROC
Multimodal backbone	$\mathbf{0.1851 \pm 0.0662}$	0.1751 ± 0.0406	2.1437 ± 0.0639	1.3721 ± 0.0360	0.5608 ± 0.0208
Soft pairwise consistency	0.1734 ± 0.0279	$\mathbf{0.2120 \pm 0.0354}$	$\mathbf{2.1348 \pm 0.0176}$	$\mathbf{1.3607 \pm 0.0182}$	$\mathbf{0.5737 \pm 0.0263}$
Hard pairwise antisymmetry	0.1053 ± 0.0458	0.1938 ± 0.0216	2.2080 ± 0.0269	1.3694 ± 0.0183	0.5575 ± 0.0058

4. Extended subgroup and robustness analyses

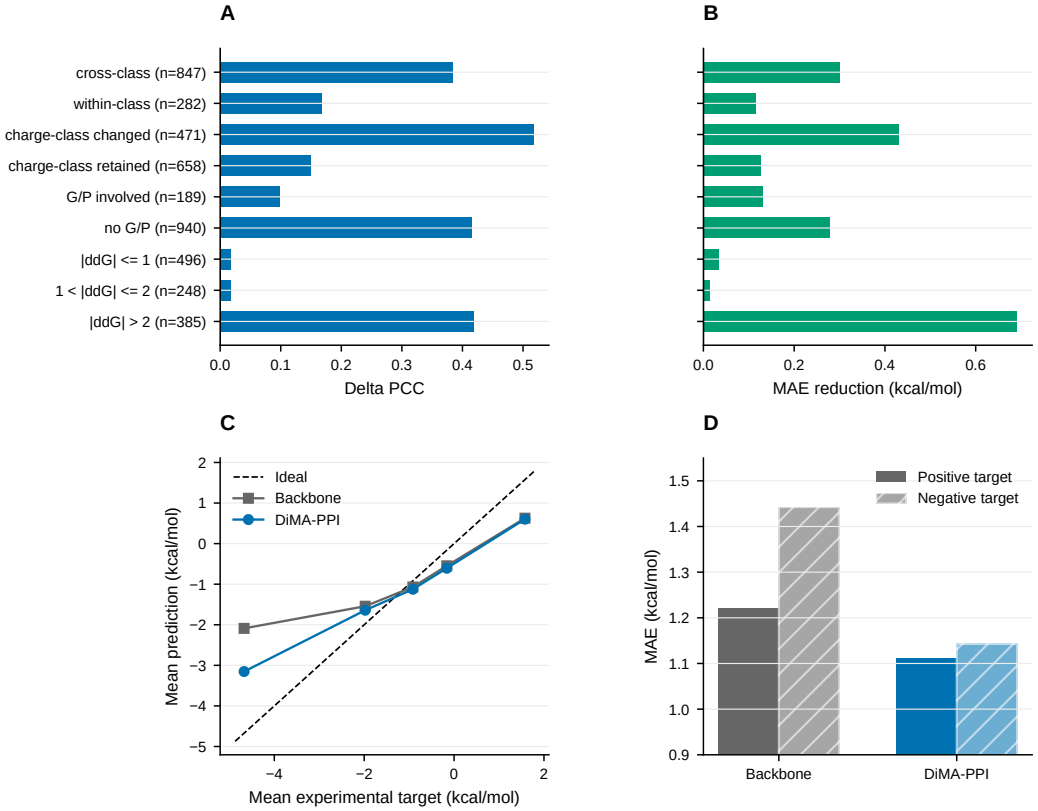


Figure S1: Complete exploratory subgroup and directional-bias audit. (A,B) PCC gain and MAE reduction across mutation chemistry and effect-size groups. (C) Calibration across experimental-target quintiles. (D) MAE for positive and negative experimental targets. These analyses were conducted after the primary evaluation and were not used for model selection.

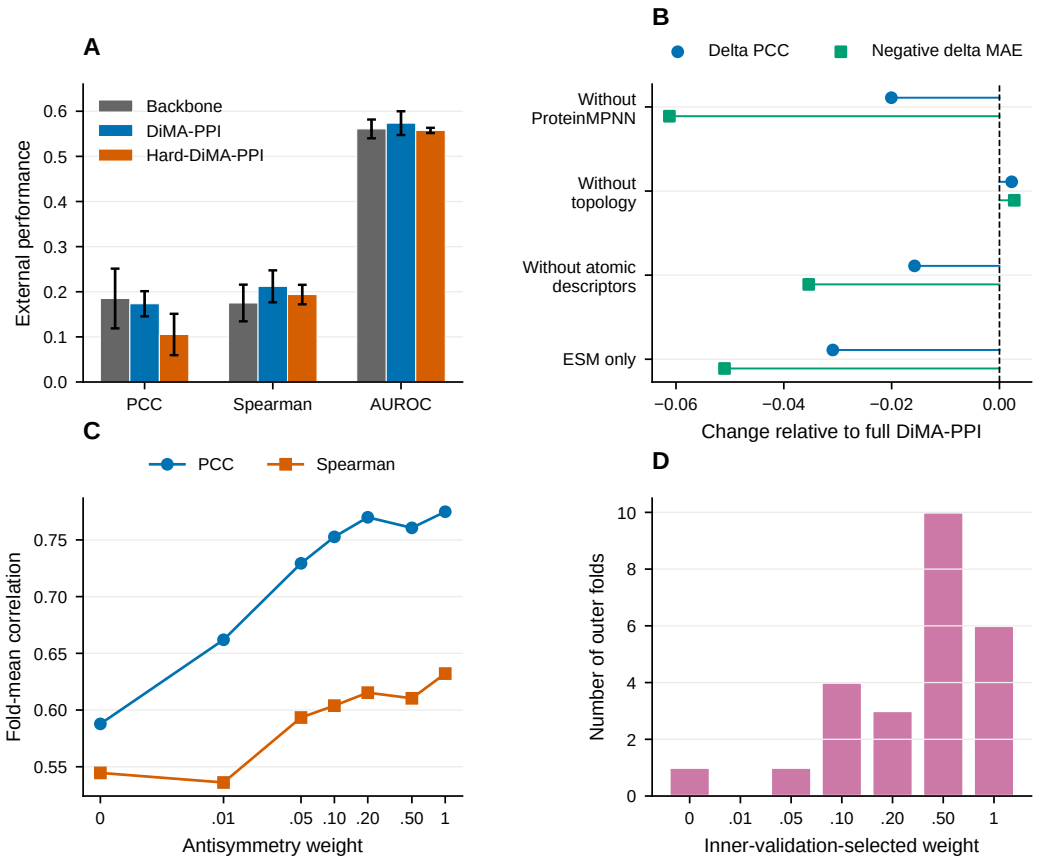


Figure S2: External transfer and secondary robustness analyses for the historical pairwise models. (A) Sequence-decontaminated AbBind transfer; no model difference was statistically supported. (B) Input-module changes relative to the full soft-consistency model. (C) Exploratory seed-43 PCC and Spearman sensitivity to the antisymmetry weight. (D) Weights selected by mutation-level inner validation across 25 outer-fold/seed combinations. Panels C and D are diagnostics rather than primary model comparisons.

These panels document historical soft-consistency experiments and are not part of the primary state-potential analysis. Mutation-level nested selection chose weights in $[0.1, 1.0]$ for 23 of 25 outer folds and yielded PCC 0.761 ± 0.009; outer-test folds were never used for selection. The fixed $\lambda_{\text{anti}} = 0.20$ result (0.759 ± 0.012 PCC) followed inspection of the exploratory seed-43 curve and is retained only as a mechanism diagnostic, not as a prospectively selected main result. A separate complex-grouped nested audit yielded PCC 0.671 ± 0.082, illustrating sensitivity to inner-validation composition. The proposed state-potential model uses neither λ_{anti} nor nested

selection of a consistency weight.

5. Case-study values

Table S5: All 16 held-out substitutions at BPTI K15 in trypsin–BPTI (PDB 2FTL). Values are five-seed ensemble predictions in kcal mol^{-1}. The site was selected post hoc as the largest per-site MAE improvement among held-out interface sites with at least five substitutions.

Mutation	Experiment	Backbone	Soft pairwise
K15A	−10.16	5.00	−6.54
K15D	−11.50	4.36	−6.95
K15E	−9.42	2.82	−6.77
K15F	−7.03	7.04	−5.44
K15G	−12.35	1.07	−7.45
K15H	−8.78	7.08	−5.78
K15I	−11.17	6.81	−6.19
K15L	−8.86	8.20	−4.92
K15M	−7.69	7.76	−5.24
K15N	−8.03	6.79	−6.10
K15Q	−9.37	6.99	−5.97
K15S	−7.72	6.22	−6.49
K15T	−10.60	6.05	−6.34
K15V	−11.76	7.36	−6.22
K15W	−8.66	8.10	−5.16
K15Y	−6.86	7.74	−5.21

6. c-MET laboratory panel

Table S6: Batch-normalized measurements and structure-free predictions for the 1-7 anti-c-MET nanobody panel. Predictions are mean ± standard deviation over five M1707-trained seeds. Positive $y_{\mathrm{DSSA}} = -RT\ln(K_{D,\mathrm{mut}}/K_{D,\mathrm{WT}})$ denotes affinity enhancement. †YWR reverses effect direction between batches and is excluded from primary metrics.

Variant	Joint substitutions	K_D^{std} (nM)	y_{DSSA}	Unconstrained	Soft	Hard
HWR	D29H/T51W/A103R	222.4	+0.387	-0.087 ± 0.324	$+1.109 \pm 0.811$	$+2.141 \pm 0.477$
YWR†	D29Y/T51W/A103R	531.8	−0.150	$+0.218 \pm 0.391$	$+1.555 \pm 0.829$	$+2.864 \pm 0.546$
HWK	D29H/T51W/A103K	82.6	+0.998	-0.026 ± 0.383	$+1.141 \pm 0.802$	$+2.160 \pm 0.492$
YWK	D29Y/T51W/A103K	91.8	+0.933	$+0.253 \pm 0.406$	$+1.587 \pm 0.818$	$+2.873 \pm 0.558$
HWY	D29H/T51W/A103Y	90.7	+0.940	$+0.442 \pm 0.337$	$+1.765 \pm 0.778$	$+3.244 \pm 0.542$
SWK	D29S/T51W/A103K	139.7	+0.674	-0.336 ± 0.371	$+0.727 \pm 0.719$	$+1.591 \pm 0.377$
SWQ	D29S/T51W/A103Q	126.3	+0.736	-0.295 ± 0.282	$+0.653 \pm 0.618$	$+1.566 \pm 0.473$
YWH	D29Y/T51W/A103H	392.6	+0.037	$+0.235 \pm 0.336$	$+1.530 \pm 0.788$	$+2.887 \pm 0.523$
NWR	D29N/T51W/A103R	1203.4	−0.653	-0.380 ± 0.410	$+0.858 \pm 0.800$	$+1.671 \pm 0.391$

HWR is directionally consistent across its two batches ($\Delta\Delta G_{K_D} = -0.502$ and -0.273 kcal/mol), whereas YWR changes sign (-0.202 and $+0.501$ kcal/mol). The pooled YWR value is therefore a low-confidence summary rather than a reliable thermodynamic label.